CodeAlpha

Data Analytics Internship

Project Documentation

Name: Panashe Guze **Domain:** Data Analytics **Tasks Completed:** Task 1, Task 2, Task 3

Tools & Technologies Used

- **Python 3.13**
- **BeautifulSoup4** — Web Scraping
- **Requests** — HTTP requests
- **Pandas** — Data Analysis
- **Matplotlib** — Data Visualization
- **Seaborn** — Statistical Visualization
- **CSV** — Data Storage

Task 1: Web Scraping

Objective

Extract structured datasets from public websites using Python libraries.

What I Did

Built three web scrapers from scratch:

1. Books Scraper (scraper.py)

- Website: books.toscrape.com
- Data collected: Title, Price, Rating
- Pages scraped: 50 pages
- Dataset size: 1000 rows

2. Hockey Teams Scraper (hockey.py)

- Website: scrapethissite.com
- Data collected: Team Name, Year, Wins, Losses
- Pages scraped: 24 pages
- Dataset size: 582 rows

3. Wikipedia Movies Scraper (movies.py)

- Website: Wikipedia — List of Highest Grossing Films
- Data collected: Title, Worldwide Gross, Year
- Dataset size: 50 rows

4. AJAX/JSON Scraper (ajax_movies.py)

- Website: scrapethissite.com/ajax
- Data collected: Title, Year, Awards, Nominations, Best Picture
- Years scraped: 2011-2015
- Dataset size: 75 rows

Key Learnings

- Inspecting HTML structure to identify data tags and classes
- Handling encoding issues with UTF-8
- Using headers to bypass website restrictions
- Pagination handling across multiple pages
- Difference between static HTML scraping and AJAX/JSON scraping

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Task 2: Exploratory Data Analysis (EDA)

Objective

Explore the hockey dataset to find patterns, trends and anomalies.

Dataset Used

Hockey Teams dataset — 582 rows, 4 columns (Team Name, Year, Wins, Losses)

Data Quality Check

- Zero missing values
- Zero duplicate rows
- Correct data types across all columns

Key Questions Asked & Findings

1. Which team had the best single season?

- Detroit Red Wings (1995) — 62 wins, 13 losses, 82.67% win rate

2. Which team had the worst single season?

- San Jose Sharks (1992) — 11 wins, 71 losses, 12.5% win rate

3. Which team dominated overall?

- Detroit Red Wings — 986 total wins across all seasons
- Consistent top performers: New Jersey Devils, Pittsburgh Penguins

4. What anomalies were discovered?

- 2004 season completely missing — **NHL lockout, entire season cancelled**
- 1994 had unusually low average wins — **player strike, shortened season**
- Post-2005 wins jumped to exactly 41 average — **new NHL rules introduced**

5. Did wins and losses correlate?

- Wins vs Losses: **-0.67** — strong negative correlation
- Wins vs Win%: **0.91** — very strong positive correlation

New Feature Created

- Calculated **Win Percentage** = Wins / (Wins + Losses) × 100

Task 3: Data Visualization

Objective

Transform raw data into visual formats that reveal insights clearly.

Charts Created

1. Bar Chart — Top 10 Teams by Total Wins

- Shows Detroit Red Wings dominance clearly
- Reveals consistent top performing franchises

2. Line Chart — Average Wins Per Year (1990-2011)

- Clearly shows 2004 gap (lockout season)
- Shows 1994 dip (player strike)
- Shows post-2005 rule change impact

3. Pie Chart — Overall Win/Loss Distribution

- Shows overall league win/loss ratio
- 53.3% wins vs 46.7% losses across all teams

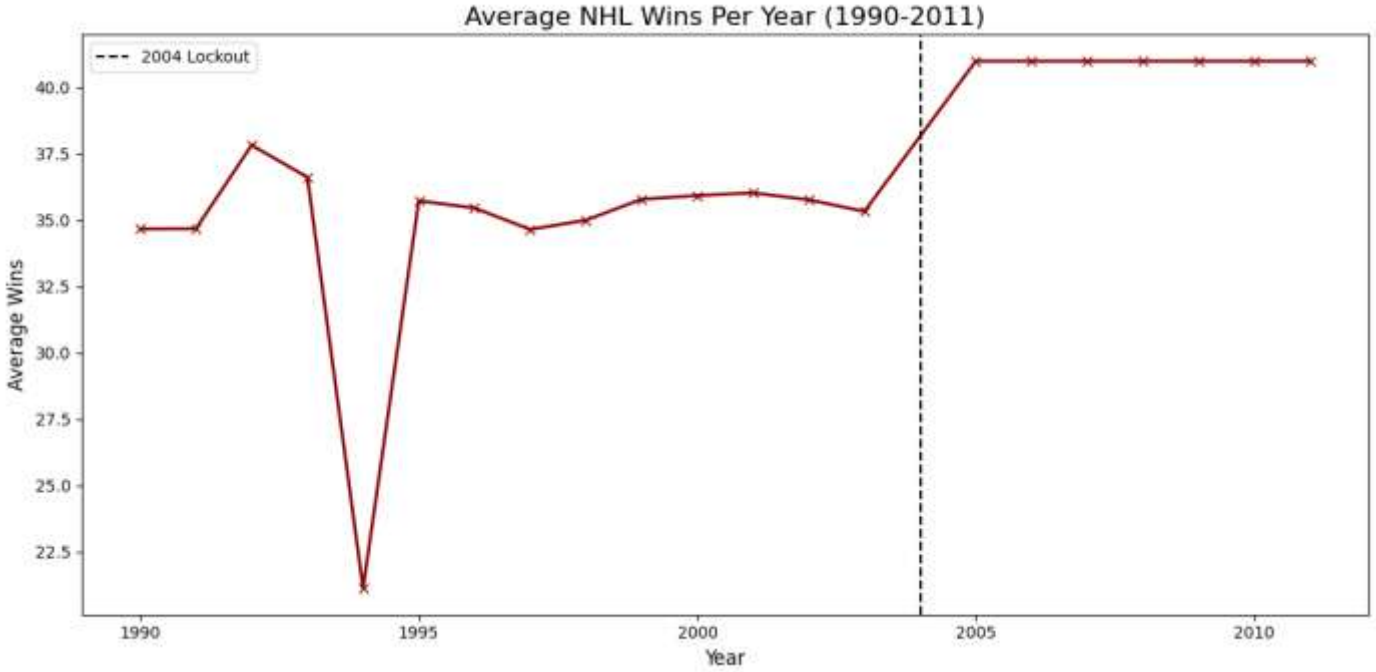
4. Heatmap — Correlation Matrix

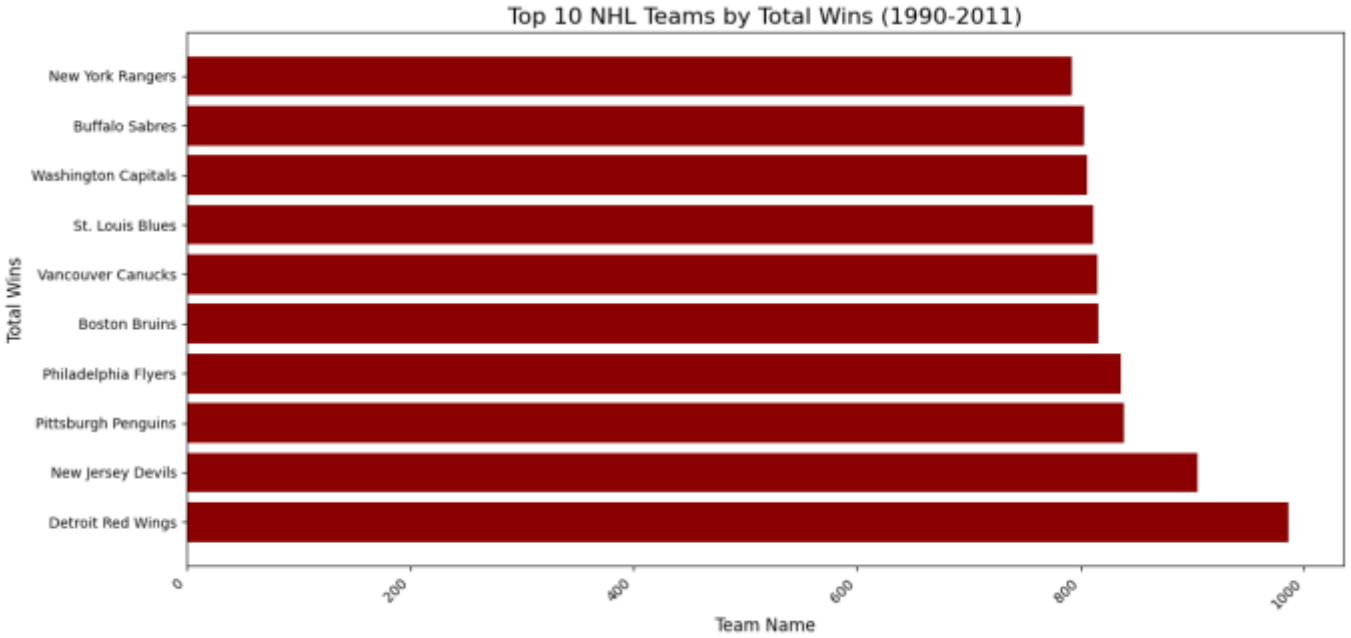
- Visual representation of relationships between Wins, Losses and Win%
- Confirms strong negative correlation between wins and losses

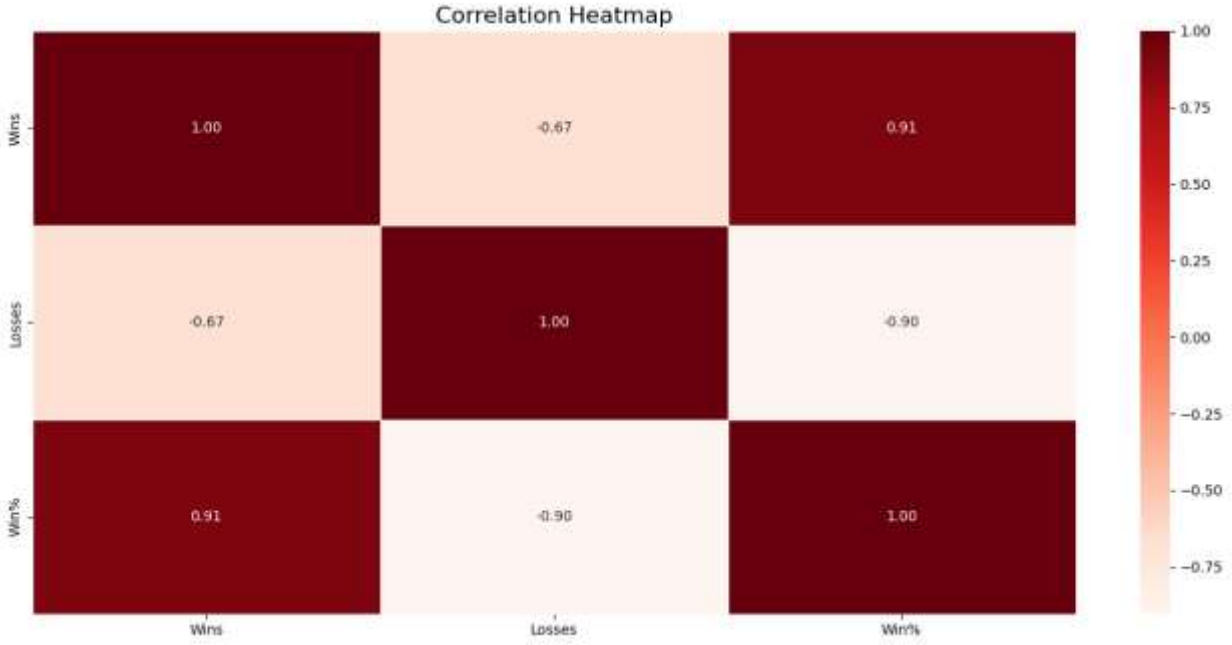
Overall Insights

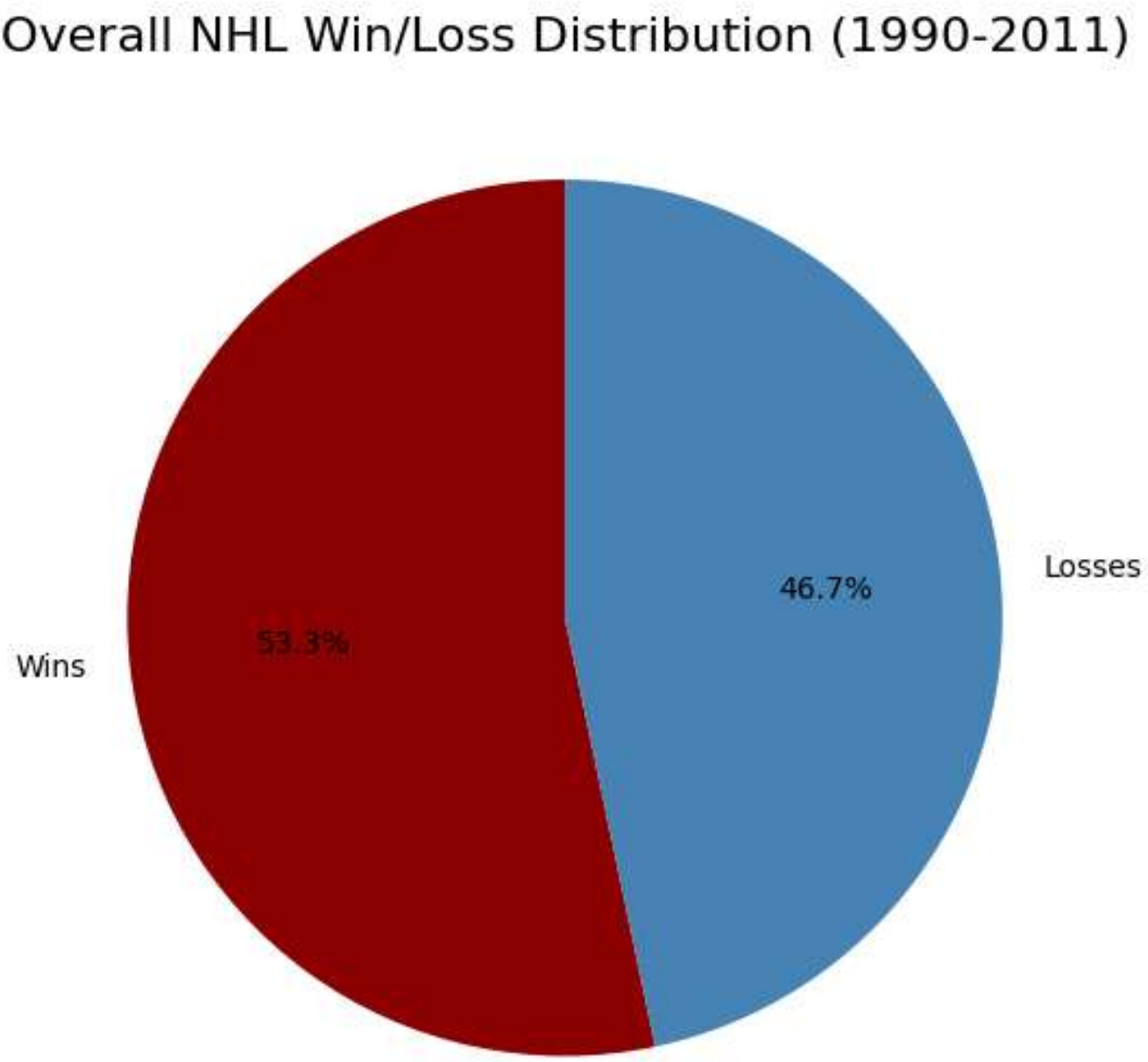
Through this project I discovered that:

1. Detroit Red Wings were the most dominant NHL team from 1990-2011
2. The NHL grew from 21 to 30 teams over this period
3. External events (strikes, lockouts) are clearly visible in the data
4. Win percentage is a better measure of team quality than raw wins alone
5. AJAX/JSON scraping is more efficient than HTML scraping when available


Average NHL Wins Per Year (1990-2011)
2004 Lockout
Average Wins
Year
40.0
37.5
35.0
32.5
30.0
27.5
25.0
22.5
1990
1995
2000
2005
2010


Top 10 NHL Teams by Total Wins (1990-2011)
New York Rangers
Buffalo Sabres
Washington Capitals
St. Louis Blues
Vancouver Canucks
Boston Bruins
Philadelphia Flyers
Pittsburgh Penguins
New Jersey Devils
Detroit Red Wings
Total Wins
Team Name
0
200
400
600
800
1000


Correlation Heatmap
Wins
Losses
Win%
1.00
-0.67
0.91
-0.67
1.00
-0.90
0.91
-0.90
1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75


Overall NHL Win/Loss Distribution (1990-2011)
Losses
46.7%
53.3%
Wins